

Cooling Hot Foods (Two-Stage Cooling)

Illustrative content — verify against your local health code.

Purpose & Scope

- Document owner: Samuel Okafor (Sous Chef, Prep)
- This SOP is for the Prep station. It applies to soups, stocks, sauces, braises, rice, beans, and other hot TCS foods we cool for later service.
- Goal: Move food through the danger zone fast and safely, without sacrificing quality.

Limits

Stage	From	To	Time limit
1	135°F	70°F	2 hours
2	70°F	41°F	a further 4 hours (6 h in total)

If stage 1 is missed, **reheat to 165°F once and restart cooling**, or discard.

Tools and Setup

- Shallow hotel pans: 2 in for liquids; 3 in for thick foods.
- Sheet trays with racks and parchment (for rice and shallow spread items).
- Ice bath station: food-safe bus tubs, ice, cold water, salt (optional to drop temp faster).
- Ice wands (frozen paddles) for stockpots and buckets; keep backups frozen.
- Blast chiller, hard chill cycle; sanitize probe thermometer before and after use.
- Calibrated digital probe thermometers; alcohol wipes; sanitizer bucket and towels.
- Labels and tape for product name, date, and time started cooling.
- Space in walk-in cleared before you start; wire racks for airflow.

Methods (use at least two together)

1. **Divide** — portion into hotel pans no deeper than **2 in** (liquids) or **3 in** (thick foods).
2. **Ice bath** — set the pans in ice water that reaches the level of the food; stir every 15 min.
3. **Ice wands** — frozen paddles in stockpots and buckets; replace as they melt.

4. **Blast chiller** — load at 135°F or below, probe in the thickest item, run the *hard chill* cycle.

Notes:

- Combine methods. Example: Divide + ice bath + stir; or Divide + blast chill + occasional stirring.
- Stirring moves hot center to surface; use clean, sanitized tools each time.
- Keep lids ajar or uncovered during cooling until food is below 41°F to vent steam.

Step-by-Step Cooling Procedure

5. Verify finish temp at or above 135°F before starting cooling. Kill the heat.

6. Divide immediately:

- Liquids (soups, stocks, sauces): transfer to 2 in hotel pans.
- Thick foods (chili, stews, braises): 3 in hotel pans max. Spread level.
- Rice/starches: spread in a thin layer (≤ 2 in) on sheet trays; move to cooling right away.

7. Start two methods at once:

- Option A: Set pans in an ice bath up to food level. Add a pinch of salt to the bath to drop temp faster. Stir every 15 min.
- Option B: Insert sanitized ice wands in deep containers; move the wand around gently as you stir. Swap wands as they melt.
- Option C: Load the blast chiller at 135°F or below. Insert the chiller probe into the thickest item. Run the hard chill cycle.

8. Maximize airflow:

- Do not stack hot pans. Leave space between pans. Use speed racks.
- In the walk-in or chiller, avoid plastic wrap or tight lids until food is below 41°F.

9. Monitor temps:

- Temp at 0 h (immediately as cooling starts).
- Temp at 2 h (must be 70°F or lower).
- Temp at 6 h (must be 41°F or lower).

10. Finish and store:

- Once $\leq 41^\circ\text{F}$, cover, label, and store on assigned shelf.
- Keep raw/ready-to-eat separation. Do not store cooling food above raw proteins.

Never

- Never put a full stockpot into the walk-in to cool.
- Never cover tightly until the food is below 41°F.

Logging

Record the temperature at **0 h, 2 h and 6 h** on the cooling log (clipboard by the blast chiller). Missing entries = discard.

Rice and starches

Cooked rice follows the same limits. Spread in a thin layer (≤ 2 in) on sheet trays and start cooling immediately — never hold cooked rice at room temperature.

Troubleshooting and What To Do If

- Stage 1 miss (above 70°F at 2 h):
 - If safe and quality permits, return to a pot, reheat to 165°F once and restart cooling using two methods. If not possible, discard.
- Too slow in ice bath:
 - Add more ice. Drain and refresh bath so ice water reaches food level. Stir more often. Split into more pans.
- Blast chiller not dropping fast:
 - Reduce load; don't overcrowd. Use shallower pans. Verify probe is in the thickest point. Combine with ice bath first, then finish in chiller.
- Thick foods still hot in the middle:
 - Increase surface area: transfer to extra pans; use ice wands while stirring. Fold in a small amount of ice if the recipe allows (soups/sauces).
- Log gaps or unreadable entries:
 - If you can't verify 0 h, 2 h, and 6 h with recorded temps, discard.
- Probe reads oddly:
 - Re-sanitize and recheck in a different spot. Calibrate if needed. When in doubt, use a second thermometer.

Food Safety and Quality Notes

- Keep hands, tools, and pans sanitized to avoid recontamination during cooling.

- Do not cool at room temp. Cooling must start immediately with active methods.
- Label every pan with product name and time cooling started. Move finished ($\leq 41^{\circ}\text{F}$) items to their final shelf promptly.
- Reheat only once if stage 1 is missed. Do not reheat multiple times to “save” a batch.
- Texture and fat: Skim fat caps after cooling; don’t let a thick fat layer insulate heat during cooling. Stir to break surface film.